THE LOG-SINE CONSTANT C_g AS THE SHARP CONSTANT IN THE RAMSEY CONTRAST OF A THERMAL CAVITY MODE

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ABSTRACT. A two-level probe coupled dispersively to a cavity mode in a thermal state with mean occupation $\bar{n}$ has the Ramsey contrast $V(u) = (1 + 4\bar{n}(\bar{n}+1)\sin^2 \pi u)^{-1/2}$, where u is the delay in units of the period of the dispersive phase. The contrast returns to 1 at every integer u. We study its geometric mean after the first return with a power weight, $\mathcal{V}_s = \exp\left(s \int_1^\infty \log V(u)\, u^{-s-1} du\right)$, $s > 0$. The main result is a two-sided bound that holds at every temperature: $p_0 < \mathcal{V}_s < e^{-sG(s)}\, p_0$, where $p_0 = 1/(\bar{n}+1)$ is the probability that the mode is empty and $G(s) = \int_1^\infty \log|2\sin \pi u|\, u^{-s-1} du$. Both constants are the best possible. For the weight u^{-2} the upper constant is $e^{C_g}/2 = 1.05195\ldots$, with $C_g = -\int_1^\infty \log|\sin \pi u|\, u^{-2} du = 0.743794\ldots$ Hence the geometric contrast after the first return equals the vacuum probability within 5.2 per cent at every temperature, and it brackets the occupation: $1/\mathcal{V}_1 - 1 < \bar{n} < e^{C_g}/(2\mathcal{V}_1) - 1$. Four further results are proved. The expansion $\mathcal{V}_1 = \frac{1}{2} e^{C_g}\, p_0\, (1-p_0)^{\pi^2/12 - 3/4} e^R$ holds with an explicit bound for R. The integral of $\log V$ over all delays with the weight u^{-2} equals $-\pi^2 \bar{n}$. The product of the contrasts at the harmonic delays N/k, $1 \le k < N$, decays as $\mathcal{V}_1^N$. The companion transform A, which collects the delays before the first return of $\log|2\sin \pi u|$ relative to its short-time form $\log(2\pi u)$, satisfies $G(s) - A(s) = \frac{\pi}{s} \cot(\frac{\pi s}{2})\, \zeta(s)$, so that in the strip $0 < \operatorname{Re} s < 2$ the non-real solutions of $G = A$ are the zeros of the Riemann zeta function. The functions G and A are shown with their computed zeros.

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1. INTRODUCTION

In a superconducting quantum processor each qubit is read out through a microwave resonator, and a few thermal photons left in that resonator are enough to dephase the qubit [8, 10]. The interaction responsible is dispersive: the transition frequency of the qubit shifts by a fixed amount for every photon in the mode [1]. The same interaction turns the qubit into a probe of the photon statistics. The problem considered here is how the thermal occupation of a mode is read from the coherence of a two-level probe.

Two families of methods are established. The first works in the frequency domain. When the shift per photon exceeds all linewidths, the transition of the qubit splits into one line for each

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photon number. This was predicted in [5] and observed in [9], where the strengths of the lines distinguished a coherent field from a thermal one. A pulse that is selective on one line maps a chosen photon number onto the state of the qubit [6], and Rabi oscillations driven on the separate lines are used to determine the occupation of the mode [10]. The second family works in the time domain. The coherence of the probe after a delay is the characteristic function of the photon number. For a thermal mode in contact with a bath it is known in closed form: the result is due to Dykman and Krivoglaz [4] and was derived again, including a drive, by Clerk and Utami [2]. Its long-time limit gives the dephasing rate that is used to interpret coherence times [8]. More generally, the coherence of a probe is a partition function of the system to which it is coupled, continued to a complex parameter [11], and such functions and their zeros have been measured [7].

These closed forms give the contrast at each delay, as a function of the delay and of the occupation together. In a mode without damping the contrast is periodic in the delay and returns to 1 at every multiple of the period π/χ of the dispersive phase, where χ is the shift per photon; we call these instants the returns. Between two returns the contrast falls to $1/(2\bar{n}+1)$, where $\bar{n}$ is the mean occupation.

The objective of this note is to determine how the contrast accumulated after the first return depends on the temperature of the mode, when it is averaged geometrically over the delay with a power weight, and to find the best constants in that dependence.

The answer is Theorem 4.1. For every exponent $s > 0$ of the weight and every temperature, the geometric contrast $\mathcal{V}_s$ lies strictly between the probability p_0 that the mode is empty and $e^{-sG(s)}p_0$, and neither constant can be improved. Here G is the transform (3.12) of $\log|2\sin\pi u|$ on $[1,\infty)$. For the weight u^{-2} the upper constant is $e^{C_g}/2$, where C_g is the log-sine constant (3.14). Corollary 4.2 turns the bound into a bracket for the occupation. Theorem 4.3 gives the expansion of $\mathcal{V}_1$ at high temperature with an explicit remainder. Proposition 5.1 shows that the integral of $\log V$ over all delays with the weight u^{-2} is exactly $-\pi^2\bar{n}$. Proposition 5.2 shows that the product of the contrasts at the harmonic delays N/k decays as $\mathcal{V}_1^N$. Section 6 treats the function G at complex exponents together with the companion transform A of (6.2), which collects the delays before the first return, proves the identity (6.4) that links them to the Riemann zeta function, and shows both functions with their computed zeros.

All statements are theorems about the ideal model (2.1): a single mode without damping, with a dispersive shift that is exactly linear in the photon number. The effect of damping on the contrast is contained in the closed form of [4, 2] and is not treated here. The proofs are elementary and self-contained. Three classical formulas are quoted from [3]: the product for the sine, the partial fractions of the cotangent and the Dirichlet integral. The figures and tables are computed in double precision or in multiple precision arithmetic and illustrate the statements. Table 1 lists the notation.

2. The model and the contrast

Let $\hat{a}$ be the annihilation operator of the mode and $\hat{\sigma}_z$ the Pauli operator of the probe, and let

$$\hat{H} = \omega_c\,\hat{a}^\dagger\hat{a} + \frac{\omega_p}{2}\,\hat{\sigma}_z + \chi\,\hat{a}^\dagger\hat{a}\,\hat{\sigma}_z, \tag{2.1}$$

in units in which the reduced Planck constant is 1, with real ω_c, ω_p and $\chi > 0$. The mode is in the thermal state of mean occupation $\bar{n} > 0$, in which the photon number $n = 0, 1, 2, \ldots$ has probability

$$p_n = (1-q)\,q^n, \qquad q = \frac{\bar{n}}{\bar{n}+1}, \qquad p_0 = 1-q = \frac{1}{\bar{n}+1}. \tag{2.2}$$

The probe is prepared in an equal superposition of the two eigenstates of $\hat{\sigma}_z$. The Hamiltonian is diagonal in the photon number, and the two eigenstates of $\hat{\sigma}_z$ with n photons have energies that differ by $\omega_p + 2\chi n$. After a time t the off-diagonal element of the state of the probe is therefore its initial value multiplied by $e^{-i\omega_p t}\sum_{n\ge0} p_n e^{-2i\chi t n}$. We measure the delay in units of the period π/χ

TABLE 1. Notation.

symbol	meaning
$\hat{a}$, $\hat{\sigma}_z$, $\hat{H}$	annihilation operator of the mode, Pauli operator of the probe, Hamiltonian (2.1)
ω_c, ω_p, χ	frequencies of the mode and of the probe, dispersive shift per photon
t, $u = \chi t/\pi$	delay, and delay in units of the period π/χ
n, p_n	photon number and its probability (2.2)
$\bar{n}$, $q = \bar{n}/(\bar{n}+1)$, $p_0 = 1-q$	mean occupation, ratio of the thermal distribution, vacuum probability
$\Lambda = 4\bar{n}(\bar{n}+1)$, $L = \log(2\bar{n}+1)$, $\tau = \log(1/q)$	auxiliary thermal parameters
$D(u)$, $V(u) = \lvert D(u)\rvert$	coherence and contrast of the probe, (2.3)
$\Phi_q(u)$, $h(u)$	$\log\lvert 1-qe^{2\pi iu}\rvert$ and $\log\lvert 2\sin\pi u\rvert$, (2.5) and (3.12)
$s = \sigma + i\theta$	exponent of the weight u^{-s-1}
$c_m(s)$, $m \geq 1$	coefficients (3.1)
$G(s)$, $G_q(s)$, $A(s)$	transforms (3.12), (3.13) and (6.2)
C_g	log-sine constant (3.14)
$\mathcal{V}_s$	geometric contrast of order s after the first return, (4.1)
$R(\tau)$	remainder of the expansion (4.6)
$\mathcal{P}(N)$, $u_k = N/k$, I_k	product of contrasts at harmonic delays, (5.3), the delays and the intervals between them
ζ	Riemann zeta function, $\zeta(s) = \sum_{j\geq 1} j^{-s}$ for $\sigma > 1$
$J(s)$, K_σ	integrals (6.8) and (6.7)
m, j, k, N, k_0, M	summation indices and integers
r, α, λ, δ, Y, x, y, v, w, B_1, B_2; z	auxiliary real variables and constants in the proofs; a complex variable

of the second factor, $u = \chi t/\pi$, and put

$$D(u) = \sum_{n\geq 0} p_n\, e^{-2\pi i n u}, \qquad V(u) = |D(u)|, \qquad u \in \mathbb{R}. \tag{2.3}$$

The function D is the coherence of the probe and V is its contrast, the quantity measured in a Ramsey experiment with delay t. The function D is the characteristic function of the geometric distribution (2.2), and the closed form below is the case without damping of the result of [4, 2]. We include its short proof.

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Proposition 2.1. *For every real u,*

$$D(u) = \frac{1-q}{1-q\,e^{-2\pi iu}}, \qquad V(u) = \frac{1}{\sqrt{1+\Lambda\sin^2(\pi u)}}, \qquad \Lambda = 4\bar{n}(\bar{n}+1), \tag{2.4}$$

and

$$\log V(u) = \log p_0 - \Phi_q(u), \qquad \Phi_q(u) = \log\left|1-q\,e^{2\pi iu}\right| = -\sum_{m\geq 1}\frac{q^m}{m}\cos(2\pi m u). \tag{2.5}$$

The contrast has period 1, satisfies $1/(2\bar{n}+1) \leq V(u) \leq 1$, and $V(u) = 1$ exactly when u is an integer.

Proof. The sum in (2.3) is a geometric series of ratio $q\,e^{-2\pi iu}$, of modulus $q < 1$. This gives D. For real α, $|1-q\,e^{-i\alpha}|^2 = 1 - 2q\cos\alpha + q^2 = (1-q)^2 + 4q\sin^2(\alpha/2)$. With $\alpha = 2\pi u$ and $4q/(1-q)^2 = 4\bar{n}(\bar{n}+1)$ this gives V. Since $1-q\,e^{-2\pi iu}$ and $1-q\,e^{2\pi iu}$ are complex conjugates, $\log V(u) = \log(1-q) - \log|1-q\,e^{2\pi iu}|$. The power series of the principal logarithm, $\log(1-z) =$

$-\sum_{m\geq 1} z^m/m$ for complex z with $|z|<1$, taken at $z = q\,e^{2\pi i u}$, has real part $-\sum_m \frac{q^m}{m}\cos(2\pi m u)$. The term $\Lambda \sin^2(\pi u)$ has period 1, vanishes exactly at the integers and is at most Λ, and $1+\Lambda = (2\bar{n}+1)^2$. □

The integers $u = 1, 2, 3, \ldots$ are the returns of the contrast. Figure 1 shows V for three temperatures.

3. The coefficients of the contrast and the transform after the first return

Throughout, $s = \sigma + i\theta$ is a complex number with real part σ and imaginary part θ. For an integer $m \geq 1$ and $\sigma > 0$ put

$$c_m(s) = \int_1^\infty \cos(2\pi m u)\, u^{-s-1}\, du. \tag{3.1}$$

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Lemma 3.1. *(a) For every integer $m \geq 1$ and $\sigma > 0$,*

$$c_m(s) = \frac{s+1}{4\pi^2 m^2}\Big(1 - (s+2)\int_1^\infty \cos(2\pi m u)\, u^{-s-3}\, du\Big). \tag{3.2}$$

The right side is analytic for $\sigma > -2$ and is taken as the definition of $c_m(s)$ there. It satisfies

$$|c_m(s)| \leq \frac{|s+1|}{4\pi^2 m^2}\Big(1 + \frac{|s+2|}{\sigma+2}\Big), \qquad \sigma > -2. \tag{3.3}$$

(b) For every integer $m \geq 1$ and real $s > -1$,

$$0 < c_m(s) < \frac{s+1}{2\pi^2 m^2}. \tag{3.4}$$

For real s with $-2 < s < -1$, $c_m(s) < 0$, and $c_m(-1) = 0$. (c) The coefficients at $s = 1$ satisfy

$$\sum_{m\geq 1} c_m(1) = \frac{\pi^2}{12} - \frac{3}{4}. \tag{3.5}$$

Proof. Throughout the proof m denotes an arbitrary integer with $m \geq 1$.

(a) Let $Y > 1$. Since $\sin(2\pi m) = 0$, one integration by parts gives

$$\int_1^Y \cos(2\pi m u)\, u^{-s-1} du = \frac{\sin(2\pi m Y)}{2\pi m}\, Y^{-s-1} + \frac{s+1}{2\pi m}\int_1^Y \sin(2\pi m u)\, u^{-s-2} du. \tag{3.6}$$

Since $\cos(2\pi m) = 1$, a second one gives

$$\int_1^Y \sin(2\pi m u)\, u^{-s-2} du = \frac{1}{2\pi m} - \frac{\cos(2\pi m Y)}{2\pi m}\, Y^{-s-2} - \frac{s+2}{2\pi m}\int_1^Y \cos(2\pi m u)\, u^{-s-3} du. \tag{3.7}$$

For $\sigma > 0$ let $Y \to \infty$ in (3.6) and (3.7). The terms with Y^{-s-1} and Y^{-s-2} tend to 0, and (3.2) follows. For $\sigma > -2$ the integral in (3.2) converges absolutely and locally uniformly in s, with modulus at most $\int_1^\infty u^{-\sigma-3} du = 1/(\sigma+2)$. Hence the right side is analytic and (3.3) holds.

(b) Let $s > -2$ be real. Then $(s+2)\int_1^\infty u^{-s-3} du = 1$. The function $\cos(2\pi m u)$ has modulus at most 1 and modulus less than 1 off a discrete set. So $(s+2)\int_1^\infty \cos(2\pi m u) u^{-s-3} du$ lies strictly between -1 and 1, and the bracket in (3.2) lies strictly between 0 and 2. The sign of $c_m(s)$ is that of $s+1$, and (3.4) follows for $s > -1$.

(c) Letting $Y \to \infty$ in (3.6) with $s = 1$ gives

$$c_m(1) = \frac{1}{\pi m}\int_1^\infty \sin(2\pi m u)\, u^{-3}\, du. \tag{3.8}$$

Let $0 < r < 1$. The series $\sum_m r^m/(\pi m)$ converges and u^{-3} is integrable on $[1, \infty)$, so the sum and the integral may be exchanged:

$$\sum_{m\geq 1} r^m\, c_m(1) = \int_1^\infty \Big(\frac{1}{\pi}\sum_{m\geq 1}\frac{r^m}{m}\sin(2\pi m u)\Big) u^{-3}\, du. \tag{3.9}$$

By the power series of the principal logarithm, $\sum_{m\ge1}\frac{r^m}{m}\sin(2\pi mu)$ is the imaginary part of $-\log(1-re^{2\pi iu})$, that is, minus the argument of $1-re^{2\pi iu}$. This number has real part at least $1-r>0$, so its argument lies between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, and the bracket in (3.9) has modulus at most $\frac12$. Let u be real and not an integer, and let $\alpha=2\pi(u-\lfloor u\rfloor)$, so that $0<\alpha<2\pi$. As $r\to1^-$, $1-re^{2\pi iu}$ tends to $1-e^{i\alpha}=2\sin(\alpha/2)\,e^{i(\alpha/2-\pi/2)}$, with $\sin(\alpha/2)>0$ and $-\frac{\pi}{2}<\frac{\alpha}{2}-\frac{\pi}{2}<\frac{\pi}{2}$. Its argument is $\frac{\alpha}{2}-\frac{\pi}{2}$, so the bracket in (3.9) tends to $\frac12-(u-\lfloor u\rfloor)$. By (3.4) the series $\sum_m c_m(1)$ converges absolutely, so its sum is the limit of the left side of (3.9) as $r\to1^-$. By dominated convergence on the right side,

$$\sum_{m\ge1}c_m(1)=\int_1^\infty\Big(\frac12-(u-\lfloor u\rfloor)\Big)u^{-3}\,du=\sum_{N\ge1}\int_N^{N+1}\Big(N+\frac12-u\Big)u^{-3}\,du. \tag{3.10}$$

For an integer $N\ge1$, $\int_N^{N+1}u^{-3}du=\frac{2N+1}{2N^2(N+1)^2}$ and $\int_N^{N+1}u^{-2}du=\frac{1}{N(N+1)}$. With $(2N+1)^2=4N(N+1)+1$,

$$\int_N^{N+1}\Big(N+\frac12-u\Big)u^{-3}\,du=\frac{(2N+1)^2}{4N^2(N+1)^2}-\frac{1}{N(N+1)}=\frac{1}{4N^2(N+1)^2}. \tag{3.11}$$

Finally $\frac{1}{N^2(N+1)^2}=\big(\frac1N-\frac{1}{N+1}\big)^2=\frac{1}{N^2}+\frac{1}{(N+1)^2}-\frac{2}{N(N+1)}$, and $\sum_{N\ge1}N^{-2}=\frac{\pi^2}{6}$, $\sum_{N\ge1}(N+1)^{-2}=\frac{\pi^2}{6}-1$, $\sum_{N\ge1}\frac{1}{N(N+1)}=1$. The sum over N of (3.11) is $\frac14(\frac{\pi^2}{3}-3)$. □

For real u that is not an integer put $h(u)=\log|2\sin(\pi u)|$. For $\sigma>0$ and $0\le q<1$ define

$$G(s)=\int_1^\infty h(u)\,u^{-s-1}\,du, \tag{3.12}$$

$$G_q(s)=\int_1^\infty \Phi_q(u)\,u^{-s-1}\,du, \tag{3.13}$$

with Φ_q as in (2.5). The integral (3.12) converges absolutely, because $\int_N^{N+1}|h(u)|du=\int_0^1|h(u)|du<\infty$ for every integer N and $\sum_{N\ge1}N^{-\sigma-1}<\infty$. The log-sine constant is

$$C_g=-\int_1^\infty\log|\sin(\pi u)|\,u^{-2}\,du=\log2-G(1). \tag{3.14}$$

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Lemma 3.2. *For $\sigma>0$ and $0\le q<1$,*

$$G_q(s)=-\sum_{m\ge1}\frac{q^m}{m}\,c_m(s),\qquad G(s)=\lim_{q\to1^-}G_q(s)=-\sum_{m\ge1}\frac{c_m(s)}{m}. \tag{3.15}$$

The last series converges absolutely and locally uniformly for $\sigma>-2$ and continues G analytically to that half-plane. On the real axis $G(-1)=0$, $G(s)<0$ for $s>-1$ and $G(s)>0$ for $-2<s<-1$. In particular

$$C_g=\log2+\sum_{m\ge1}\frac{c_m(1)}{m}=0.743794861839400\ldots \tag{3.16}$$

Proof. For $q<1$ the series in (2.5) is dominated by $\sum_m q^m/m$, and $u^{-\sigma-1}$ is integrable on $[1,\infty)$. Integrating term by term gives the first identity of (3.15).

For real α and $0\le r\le1$ we have $|\sin\alpha|\le|1-re^{i\alpha}|\le2$. Indeed, if $\cos\alpha\le0$ the modulus is at least 1, and if $\cos\alpha>0$ it is at least the distance $|\sin\alpha|$ from the point 1 to the line through 0 and $e^{i\alpha}$. Hence $|\Phi_q(u)|\le\log2+\big|\log|\sin2\pi u|\big|$ for all q, a function of period 1 that is integrable on a period. As for (3.12), this function times $u^{-\sigma-1}$ is integrable on $[1,\infty)$. For u not an integer, $\Phi_q(u)\to\log|1-e^{2\pi iu}|=h(u)$ as $q\to1^-$, because $|1-e^{i\alpha}|=2|\sin(\alpha/2)|$. By dominated convergence $G_q(s)\to G(s)$. By (3.3) the series $\sum_m|c_m(s)|/m$ converges, locally uniformly for $\sigma>-2$, so $\sum_m q^m c_m(s)/m\to\sum_m c_m(s)/m$ as $q\to1^-$, and the sum is analytic for $\sigma>-2$. The

signs on the real axis follow from Lemma 3.1(b), and (3.16) is (3.14) with $G(1) = -\sum_m c_m(1)/m$. The decimal value is computed from this series. □

4. The sharp bound

For real $s > 0$ the weight $s\,u^{-s-1}du$ is a probability measure on $[1,\infty)$. The geometric contrast of order s after the first return is the weighted geometric mean

$$\mathcal{V}_s = \exp\Big(s\int_1^\infty \log V(u)\,u^{-s-1}\,du\Big), \qquad s>0. \tag{4.1}$$

It depends on the temperature through $\bar n$.

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Theorem 4.1. *Let $s > 0$ be real and $\bar n > 0$. Then*

$$\log \mathcal{V}_s = \log p_0 - s\,G_q(s) = \log p_0 + s\sum_{m\ge1}\frac{q^m}{m}\,c_m(s), \qquad q=\frac{\bar n}{\bar n+1}. \tag{4.2}$$

The ratio $\mathcal{V}_s/p_0$ is a strictly increasing function of $\bar n$. It tends to 1 as $\bar n \to 0$ and to $e^{-sG(s)}$ as $\bar n\to\infty$. Consequently

$$p_0 < \mathcal{V}_s < e^{-sG(s)}\,p_0 \qquad \text{for every } \bar n>0, \tag{4.3}$$

and both constants are the best possible: if real numbers B_1, B_2 satisfy $B_1\,p_0 \le \mathcal{V}_s \le B_2\,p_0$ for every $\bar n>0$, then $B_1\le 1$ and $B_2 \ge e^{-sG(s)}$. For $s=1$,

$$p_0 < \mathcal{V}_1 < \frac{e^{C_g}}{2}\,p_0, \qquad \frac{e^{C_g}}{2} = 1.0519522055\ldots \tag{4.4}$$

Proof. By (2.5) and $s\int_1^\infty u^{-s-1}du = 1$, $s\int_1^\infty \log V(u)u^{-s-1}du = \log p_0 - s\,G_q(s)$, and (3.15) gives (4.2). By (3.4) every coefficient $c_m(s)$ is positive. The ratio q increases strictly from 0 to 1 as $\bar n$ increases from 0 to ∞. So $s\sum_m q^m c_m(s)/m$ is positive and strictly increasing in $\bar n$. It is at most $q\,s\sum_m c_m(s)/m$, which tends to 0 as $q\to0$, and it tends to $-sG(s)$ as $q\to1^-$ by (3.15). A strictly increasing function lies strictly between its limits, which is (4.3). If $B_1p_0 \le \mathcal{V}_s \le B_2p_0$ for all $\bar n$, the two limits of $\mathcal{V}_s/p_0$ give $B_1\le1$ and $B_2\ge e^{-sG(s)}$. For $s=1$, $-G(1) = C_g - \log 2$ by (3.14). □

The bound says that the geometric contrast after the first return is the vacuum probability of the mode, up to a factor that never exceeds $e^{-sG(s)}$. The values of G on the positive real axis are therefore the logarithms of the sharp constants: $e^{-sG(s)} = 1.02005,\ 1.05195,\ 1.14763,\ 1.28059$ for $s = \frac12,\ 1,\ 2,\ 3$. Figure 2 shows G on the real axis and Figure 1 shows the ratio $\mathcal{V}_1/p_0$.

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Corollary 4.2. *For every $\bar n > 0$,*

$$\frac{1}{\mathcal{V}_1} - 1 < \bar n < \frac{e^{C_g}}{2\,\mathcal{V}_1} - 1. \tag{4.5}$$

Proof. Divide (4.4) by $\mathcal{V}_1\,p_0 > 0$ and use $1/p_0 = \bar n + 1$. □

The bracket (4.5) contains no fitted quantity. Its two ends differ by 5.2 per cent of $\bar n+1$. For $\bar n = 20$, $\mathcal{V}_1 = 0.0499291$ and (4.5) reads $19.03 < \bar n < 20.07$.

At high temperature the ratio approaches its upper bound, and the approach can be made explicit.

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Theorem 4.3. *Let $\bar n > 0$ and $\tau = \log(1/q) = \log(1+1/\bar n)$. Then*

$$\log \mathcal{V}_1 = \log p_0 + C_g - \log 2 - \Big(\frac{\pi^2}{12} - \frac34\Big)\tau + R(\tau), \qquad R(\tau) > 0, \tag{4.6}$$

that is, $\mathcal{V}_1 = \frac12 e^{C_g}\,p_0\,q^{\pi^2/12-3/4}\,e^{R(\tau)}$. For $0<\tau\le1$, that is, for $\bar n \ge 1/(e-1)$,

$$R(\tau) \le \frac{\tau^2}{\pi^2}\Big(\frac52 + \frac12\log\frac{1}{\tau}\Big). \tag{4.7}$$

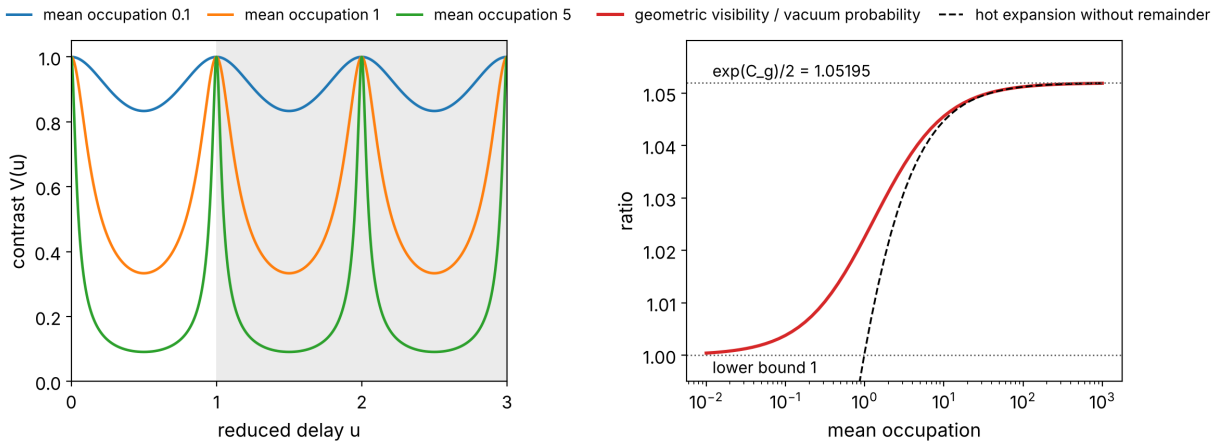


FIGURE 1. Left: the contrast $V(u)$ of (2.4) for $\bar{n} = 0.1$, 1 and 5; the shaded region $u \geq 1$ is the range of the geometric contrast (4.1). Right: the ratio $\mathcal{V}_1/p_0$ as a function of $\bar{n}$, between its sharp bounds 1 and $e^{C_g}/2$ of (4.4), and the expansion (4.6) without its remainder.

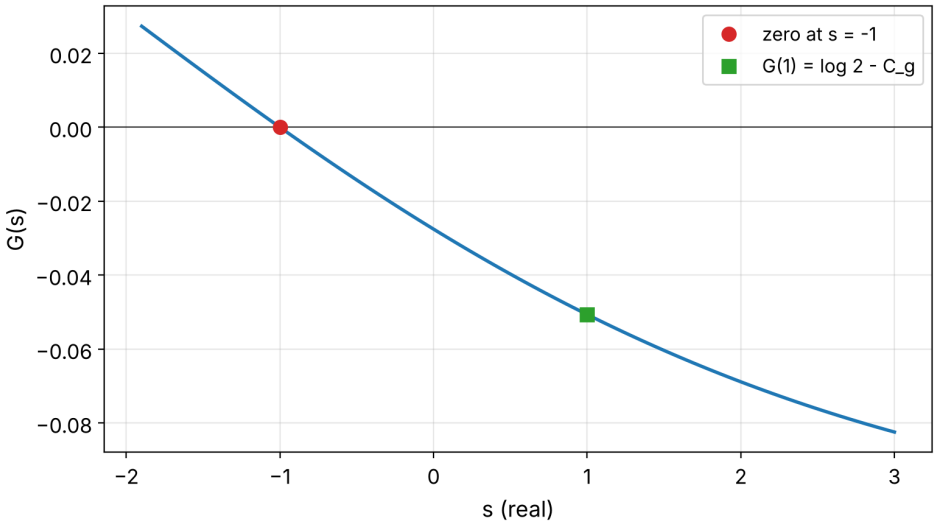


FIGURE 2. The function G on the real axis. For $s > 0$ the sharp constant of (4.3) is $e^{-sG(s)}$; the square marks $G(1) = \log 2 - C_g$.

Proof. Write $c_m = c_m(1)$ and $q^m = e^{-m\tau}$. By (4.2), (3.16) and (3.5),

$$R(\tau) = \sum_{m\geq 1} \frac{q^m}{m} c_m - \sum_{m\geq 1} \frac{c_m}{m} + \tau \sum_{m\geq 1} c_m = \sum_{m\geq 1} \frac{c_m}{m} (m\tau - 1 + e^{-m\tau}). \tag{4.8}$$

For real $y > 0$, $0 < y - 1 + e^{-y} \leq \min(y, \frac{1}{2}y^2)$. Since $c_m > 0$, this gives $R(\tau) > 0$. With $c_m < 1/(\pi^2 m^2)$ from (3.4) it also gives

$$R(\tau) \leq \frac{1}{\pi^2} \sum_{m\geq 1} \min\Big(\frac{\tau}{m^2}, \frac{\tau^2}{2m}\Big). \tag{4.9}$$

Let $0 < \tau \le 1$ and let $M = \lfloor 1/\tau \rfloor \ge 1$. Then $\sum_{m \le M} \frac{1}{m} \le 1 + \log M \le 1 + \log \frac{1}{\tau}$ and $\sum_{m > M} \frac{1}{m^2} < \frac{1}{M} \le 2\tau$, because $\lfloor y \rfloor \ge \frac{y}{2}$ for $y \ge 1$. Using the second entry of the minimum for $m \le M$ and the first for $m > M$, the sum in (4.9) is at most $\frac{\tau^2}{2}(1 + \log \frac{1}{\tau}) + 2\tau^2$. □

Table 2 gives the exact value of $\mathcal{V}_1$ from (4.2), the ratio $\mathcal{V}_1/p_0$, the remainder $R(\tau)$ and its bound (4.7).

TABLE 2. The geometric contrast $\mathcal{V}_1$ computed from (4.2), its ratio to the vacuum probability p_0, the remainder $R(\tau)$ of (4.6) and the bound (4.7), which applies when $\tau \le 1$.

$\bar{n}$	p_0	$\mathcal{V}_1$	$\mathcal{V}_1/p_0$	$R(\tau)$	bound (4.7)
0.01	0.9900990	0.9905007	1.0004057	2.84×10^{-1}	
0.1	0.9090909	0.9125237	1.0037761	1.27×10^{-1}	
1	0.5000000	0.5112376	1.0224752	2.18×10^{-2}	1.31×10^{-1}
5	0.1666667	0.1734625	1.0407747	2.53×10^{-3}	1.13×10^{-2}
20	0.0476190	0.0499291	1.0485104	2.58×10^{-4}	9.67×10^{-4}
100	0.0099010	0.0104080	1.0512094	1.47×10^{-5}	4.82×10^{-5}
1000	0.0009990	0.0010508	1.0518762	2.07×10^{-7}	6.03×10^{-7}

5. The sum rule and the harmonic delays

The first return divides the delays into the interval $0 < u < 1$ and the half-line $u \ge 1$. Over all delays together the integral of the logarithm of the contrast is linear in the occupation.

Reviewed by Claude (review 1)

Proposition 5.1. *For every $\bar{n} > 0$,*

$$\int_0^\infty \log V(u)\, u^{-2}\, du = -\pi^2\, \bar{n}, \qquad \int_0^1 \log V(u)\, u^{-2}\, du = -\pi^2\, \bar{n} - \log \mathcal{V}_1. \tag{5.1}$$

Proof. By (2.5) and $\log(1-q) = -\sum_m q^m/m$, $\log V(u) = \sum_{m \ge 1} \frac{q^m}{m}\big(\cos(2\pi m u) - 1\big)$, a series of terms that are all ≤ 0, so it may be integrated term by term. Let $\lambda > 0$ and $0 < \delta < Y$. Integration by parts gives $\int_\delta^Y (1 - \cos \lambda u) u^{-2} du = \frac{1-\cos\lambda\delta}{\delta} - \frac{1-\cos\lambda Y}{Y} + \lambda \int_\delta^Y \frac{\sin \lambda u}{u}\, du$. As $\delta \to 0$ and $Y \to \infty$ the first two terms tend to 0 and the last integral tends to $\frac{\pi}{2}$, by the classical Dirichlet integral [3, (6.2.9), (6.2.14)]. Hence $\int_0^\infty (1 - \cos \lambda u) u^{-2} du = \frac{\pi\lambda}{2}$. With $\lambda = 2\pi m$,

$$\int_0^\infty \log V(u)\, u^{-2}\, du = -\sum_{m \ge 1} \frac{q^m}{m}\, \pi^2 m = -\pi^2\, \frac{q}{1-q} = -\pi^2\, \bar{n}. \tag{5.2}$$

The second identity of (5.1) follows from $\int_1^\infty \log V(u) u^{-2} du = \log \mathcal{V}_1$, which is (4.1) with $s = 1$. □

By (4.4) the part of the sum rule that lies after the first return, $\log \mathcal{V}_1$, differs from $\log p_0$ by less than $C_g - \log 2 = 0.0506\ldots$ at every temperature. The part before the first return carries the linear term $-\pi^2 \bar{n}$.

The geometric contrast is also the rate of decay of a product. For an integer $N \ge 2$ let

$$\mathcal{P}(N) = \prod_{k=1}^{N-1} V\Big(\frac{N}{k}\Big). \tag{5.3}$$

This is the product of the contrasts measured at the harmonic delays $u_k = N/k$, that is, at the times $N\pi/(k\chi)$. It is also the contrast at the time $N\pi/\chi$ of a probe coupled to $N-1$ independent modes with dispersive shifts χ/k, $1 \le k \le N-1$, and equal occupation $\bar{n}$, because the coherences of independent modes multiply.

Reviewed by Claude (review 1)

Proposition 5.2. *Let $\bar n > 0$, $L = \log(2\bar n + 1)$ and let $N \ge 2$ be an integer. Then*

$$\Big|\frac{1}{N}\log \mathcal{P}(N) - \log \mathcal{V}_1\Big| \le L\Big(\frac{2\sqrt{2}}{\sqrt{N}} + \frac{1}{N}\Big). \tag{5.4}$$

In particular $\mathcal{P}(N)^{1/N} \to \mathcal{V}_1$ as $N \to \infty$, and by (4.4) this limit lies strictly between p_0 and $\frac{1}{2}e^{C_g}p_0$.

Proof. By Proposition 2.1 the function $\log V$ has period 1 and takes values in $[-L, 0]$. On each interval $[j, j+\frac{1}{2}]$ with integer j it decreases from 0 to $-L$, and on $[j+\frac{1}{2}, j+1]$ it increases from $-L$ to 0. Put $u_k = N/k$ for $1 \le k \le N$, so that $u_N = 1$ and $u_1 = N$, and $I_k = [u_{k+1}, u_k]$ for $1 \le k \le N-1$. The intervals I_k cover $[1, N]$ and overlap only at end points, and $\int_{I_k} u^{-2}du = \frac{k+1}{N} - \frac{k}{N} = \frac{1}{N}$. Therefore

$$\frac{1}{N}\log \mathcal{P}(N) - \int_1^N \log V(u)\, u^{-2}\, du = \sum_{k=1}^{N-1} \int_{I_k} \big(\log V(u_k) - \log V(u)\big)\, u^{-2}\, du. \tag{5.5}$$

The k-th term has modulus at most $\frac{1}{N}$ times the oscillation of $\log V$ on I_k, which is at most L and at most the total variation of $\log V$ on I_k. Let k_0 be an integer with $1 \le k_0 \le N-1$. The intervals I_k with $k > k_0$ cover $[1, u_{k_0+1}]$ and overlap only at end points, so the sum of the total variations on them is the total variation on $[1, u_{k_0+1}]$. An interval $[1, x]$ with $x \ge 1$ is covered by the intervals of monotonicity $[\frac{j}{2}, \frac{j+1}{2}]$ with $2 \le j \le \lceil 2x\rceil - 1$, at most $2x-1$ in number, on each of which the variation is at most L. So the total variation of $\log V$ on $[1, x]$ is at most $2Lx$. The right side of (5.5) is thus at most $\frac{L}{N}(k_0 + \frac{2N}{k_0+1})$ in modulus. Choose $k_0 = \lceil\sqrt{2N}\rceil - 1$. Since $N \ge 2$, $2 \le \sqrt{2N} \le N$, so $1 \le k_0 \le N-1$, and $k_0 < \sqrt{2N} \le k_0 + 1$. Then $k_0 + \frac{2N}{k_0+1} < 2\sqrt{2N}$, which gives the first term of (5.4). Finally $|\int_N^\infty \log V(u)u^{-2}du| \le \frac{L}{N}$, and $\int_1^\infty \log V(u)u^{-2}du = \log \mathcal{V}_1$. □

At zero temperature the factors of the same product, formed with $2|\sin(\pi N/k)|$ in place of the contrast, vanish whenever k divides N. The thermal factors do not vanish: by Proposition 2.1 each of them is at least $1/(2\bar n + 1)$, and a divisor k of N contributes the factor 1.

6. The transforms after and before the first return and their zeros

Theorem 4.1 uses G on the positive real axis. At complex exponents the same function describes the hot limit of the contrast against weights that oscillate in the logarithm of the delay.

Reviewed by Claude (review 1)

Proposition 6.1. *For $\sigma > 0$,*

$$\lim_{\bar n \to \infty} \int_1^\infty \log \frac{V(u)}{p_0}\, u^{-s-1}\, du = -G(s), \tag{6.1}$$

uniformly for s in compact subsets of the half-plane $\sigma > 0$.

Proof. By (2.5) the integral is $-G_q(s)$. By (3.15), $|G_q(s) - G(s)| \le \sum_{m\ge1}(1-q^m)|c_m(s)|/m$. By (3.3) the series $\sum_m |c_m(s)|/m$ converges uniformly on compact subsets of $\sigma > -2$, and each term of the bound tends to 0 as $q \to 1^-$, so the bound tends to 0 uniformly there. □

For real $s > 0$ the limit (6.1) is positive and gives the sharp constants of Theorem 4.1. For $s = \sigma + i\theta$ with $\theta \ne 0$ the weight is $u^{-\sigma-1}e^{-i\theta\log u}$. At a zero of G with $\sigma > 0$ the limit vanishes: the excess $\log(V/p_0)$ of a hot mode has no component on that weight. Table 3 lists the zeros of G computed in the rectangle $0 < \sigma < 2$, $0 < \theta \le 40$, and Figure 3 shows $|G|$ there.

The delays before the first return enter through a second function. The weight u^{-s-1} is not integrable at $u = 0$ against h, which behaves there as $\log(2\pi u)$. Subtracting this short-time form gives, for $\sigma < 2$ and $s \ne 0$,

$$A(s) = \frac{\log(2\pi)}{s} + \frac{1}{s^2} - \int_0^1 \log\frac{\sin(\pi u)}{\pi u}\, u^{-s-1}\, du. \tag{6.2}$$

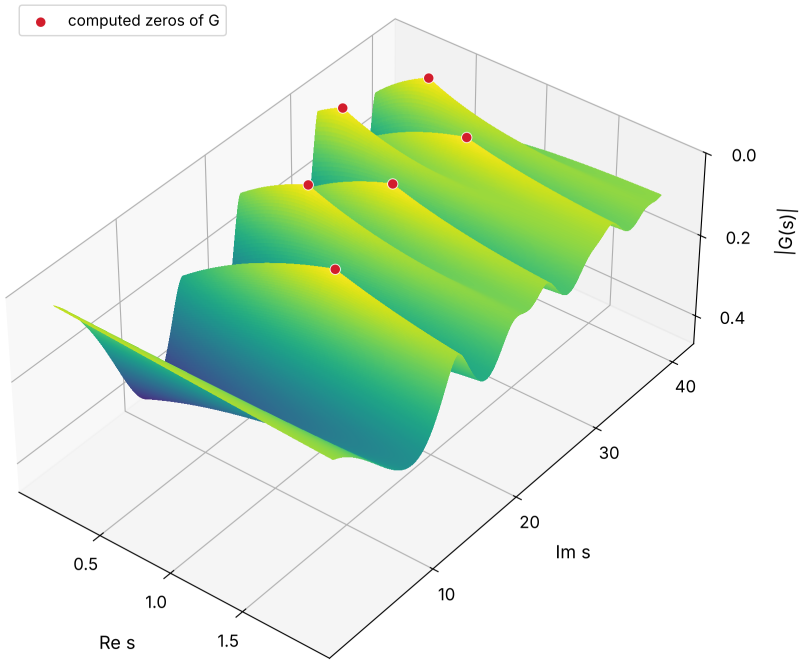


FIGURE 3. $|G(s)|$ on the rectangle $0.05 \le \operatorname{Re} s \le 1.95$, $1 \le \operatorname{Im} s \le 40$, on a linear vertical axis that is inverted, so that a zero is a peak. The red markers are the six computed zeros of G of Table 3.

The integral converges absolutely for $\sigma < 2$, because the logarithm is $O(u^2)$ as $u \to 0$ and has an integrable singularity at $u = 1$. Let ζ denote the Riemann zeta function, $\zeta(s) = \sum_{j\ge1} j^{-s}$ for $\sigma > 1$, continued analytically to the complex plane with a simple pole at $s = 1$.

Reviewed by Claude (review 1)

Proposition 6.2. *(a) For $\sigma < 2$ and $s \neq 0$,*

$$A(s) = \frac{\log(2\pi)}{s} + \frac{1}{s^2} + \sum_{k\ge1} \frac{\zeta(2k)}{k\,(2k-s)}. \tag{6.3}$$

The series converges for every complex s other than $2, 4, 6, \ldots$ and continues A to a meromorphic function on the complex plane.

(b) For $0 < \sigma < 2$,

$$G(s) - A(s) = \frac{\pi}{s} \cot\left(\frac{\pi s}{2}\right) \zeta(s), \tag{6.4}$$

where the right side is continued analytically to $s = 1$, with value $-\frac{\pi^2}{2}$.

(c) For non-real s with $0 < \sigma < 2$, $G(s) = A(s)$ if and only if $\zeta(s) = 0$.

Proof. (a) By the product for the sine [3, (4.22.1)], $\sin(\pi u) = \pi u \prod_{j\ge1}(1 - u^2/j^2)$. For $0 < u < 1$ every factor lies in $(0, 1)$ and

$$\log \frac{\sin(\pi u)}{\pi u} = \sum_{j\ge1} \log\left(1 - \frac{u^2}{j^2}\right) = -\sum_{j\ge1}\sum_{k\ge1} \frac{u^{2k}}{k\, j^{2k}} = -\sum_{k\ge1} \frac{\zeta(2k)}{k}\, u^{2k}, \tag{6.5}$$

a double series of terms of one sign. For $\sigma < 2$, $\int_0^1 u^{2k-s-1} du = \frac{1}{2k-s}$, and $\sum_k \frac{\zeta(2k)}{k(2k-\sigma)} < \infty$. Integrating (6.5) term by term in (6.2) gives (6.3). Since $\zeta(2k) \le \zeta(2)$, the terms are $O(k^{-2})$ uniformly on compact sets that avoid the points $2k$, which gives the continuation.

(b) For $\sigma > 0$, $\int_1^\infty \log(2\pi u)\, u^{-s-1} du = \frac{\log(2\pi)}{s} + \frac{1}{s^2}$. For $0 < u < 1$, $h(u) - \log(2\pi u) = \log \frac{\sin(\pi u)}{\pi u}$. Hence, for $0 < \sigma < 2$,

$$G(s) - A(s) = \int_0^\infty \big(h(u) - \log(2\pi u)\big)\, u^{-s-1}\, du. \tag{6.6}$$

By the product for the sine, $h(u) - \log(2\pi u) = \sum_{j\geq 1} \log|1 - u^2/j^2|$ for every $u > 0$ that is not an integer. For $0 < \sigma < 2$ put

$$K_\sigma = \int_0^\infty \big|\log|1 - v^2|\big|\, v^{-\sigma-1}\, dv < \infty. \tag{6.7}$$

It is finite because the logarithm is $O(v^2)$ at 0, has an integrable singularity at 1 and is $O(\log v)$ at infinity. The substitution $u = jv$ gives $\int_0^\infty \big|\log|1 - u^2/j^2|\big| u^{-\sigma-1} du = j^{-\sigma} K_\sigma$. Let $1 < \sigma < 2$. Then $\sum_j j^{-\sigma} K_\sigma < \infty$, the sum over j may be exchanged with the integral in (6.6), and

$$G(s) - A(s) = \zeta(s)\, J(s), \qquad J(s) = \int_0^\infty \log|1 - v^2|\, v^{-s-1}\, dv. \tag{6.8}$$

We compute $J(s)$ for $0 < \sigma < 2$. On $(0,1)$, $\log(1 - v^2) = -\sum_{k\geq 1} v^{2k}/k$ with terms of one sign, so $\int_0^1 \log(1 - v^2)\, v^{-s-1} dv = -\sum_k \frac{1}{k(2k-s)}$. On $(1,\infty)$ substitute $v = 1/w$: $\int_1^\infty \log(v^2 - 1)\, v^{-s-1} dv = \int_0^1 \big(\log(1 - w^2) - 2\log w\big) w^{s-1} dw = -\sum_k \frac{1}{k(2k+s)} + \frac{2}{s^2}$. Adding the two parts,

$$J(s) = \frac{2}{s^2} - \sum_{k\geq 1} \frac{1}{k}\Big(\frac{1}{2k - s} + \frac{1}{2k + s}\Big) = \frac{2}{s^2} - \sum_{k\geq 1} \frac{1}{k^2 - s^2/4}. \tag{6.9}$$

The partial fractions of the cotangent [3, (4.22.3)], at the point $\frac{\pi s}{2}$, which is not a multiple of π, read $\cot\frac{\pi s}{2} = \frac{2}{\pi s} - \frac{s}{\pi}\sum_{k\geq 1} \frac{1}{k^2 - s^2/4}$. Multiplying by $\frac{\pi}{s}$ and comparing with (6.9) gives $J(s) = \frac{\pi}{s}\cot\frac{\pi s}{2}$. This proves (6.4) for $1 < \sigma < 2$. Both sides are analytic in the strip $0 < \sigma < 2$: G by Lemma 3.2, A by (6.2), and the right side because the simple pole of ζ at $s = 1$ is cancelled by the zero of the cotangent. So (6.4) holds in the strip. Near $s = 1$, $\cot\frac{\pi s}{2} = -\frac{\pi}{2}(s - 1) + O((s-1)^3)$ and $\zeta(s) = \frac{1}{s-1} + O(1)$, which gives the value $-\frac{\pi^2}{2}$.

(c) For non-real s the number $\frac{\pi s}{2}$ is not real, so $\cot\frac{\pi s}{2}$ is finite and different from 0, and $\frac{\pi}{s} \neq 0$. By (6.4), $G(s) - A(s)$ is the product of $\zeta(s)$ and a finite non-zero number, so each of $G(s) = A(s)$ and $\zeta(s) = 0$ implies the other. □

By (6.6) the difference $G - A$ is the transform over all delays of h relative to its short-time form, and by (6.4) it is the Riemann zeta function up to an elementary factor. The function G carries the delays after the first return and A the delays before it. By Proposition 6.2(c) the two coincide at a non-real exponent of the strip exactly at the zeros of ζ.

Figures 4 and 5 show $|A|$. On the grid of Figure 4 the computed values of $|A|$ lie between 0.0721 and 2.1831, and no zero of A is found there. Near the real axis the continuation (6.3) has the poles $0, 2, 4, \ldots$ and real zeros, of which Table 3 lists those in $-3.5 < s < 9.5$. Figure 6 shows, in the rectangle of Figure 3, the six computed zeros of G and the six points with $G = A$. The latter are the points $\frac{1}{2} + i\theta$ with $\theta = 14.1347$, 21.0220, 25.0109, 30.4249, 32.9351, 37.5862, the first six zeros of ζ in the upper half-plane.

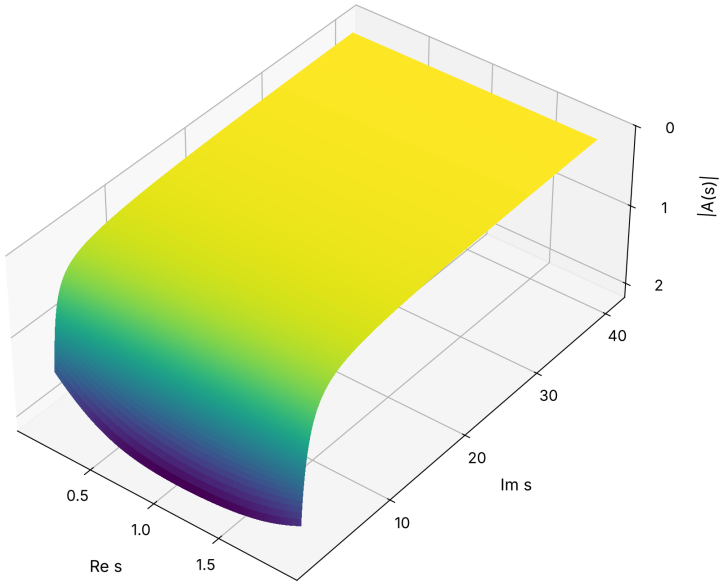


FIGURE 4. $|A(s)|$ on the rectangle of Figure 3, with the vertical axis inverted.

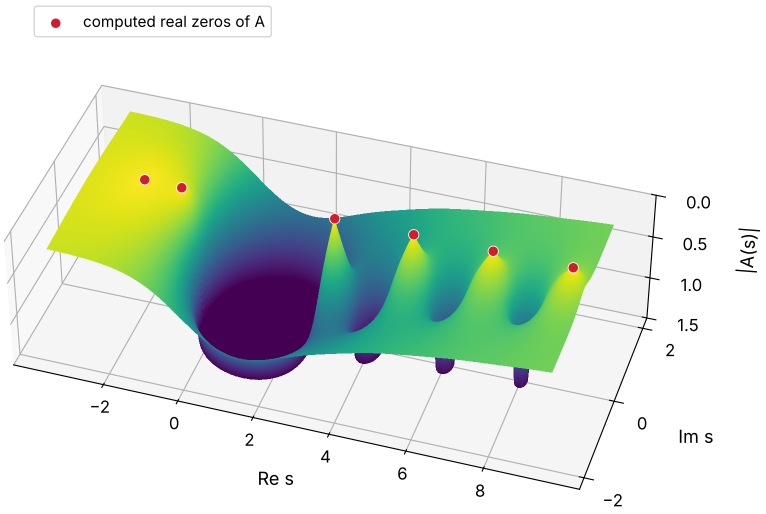


FIGURE 5. $|A(s)|$ near the real axis, computed from (6.3) and clipped above 1.5, with the vertical axis inverted. The red markers are the computed real zeros of A of Table 3; the wells are the poles $0, 2, 4, 6, 8$.

TABLE 3. Computed zeros of G in the rectangle $0 < \operatorname{Re} s < 2$, $0 < \operatorname{Im} s \le 40$, and computed real zeros of the continuation (6.3) of A in $-3.5 < s < 9.5$.

zeros of G	real zeros of A
$1.127313 + 15.001263\,i$	-2.0000
$0.533572 + 21.763255\,i$	-1.0000
$0.894731 + 25.624837\,i$	3.0428
$0.227688 + 30.975963\,i$	5.0981
$0.950507 + 33.599088\,i$	7.1478
$0.430900 + 38.054886\,i$	9.1892

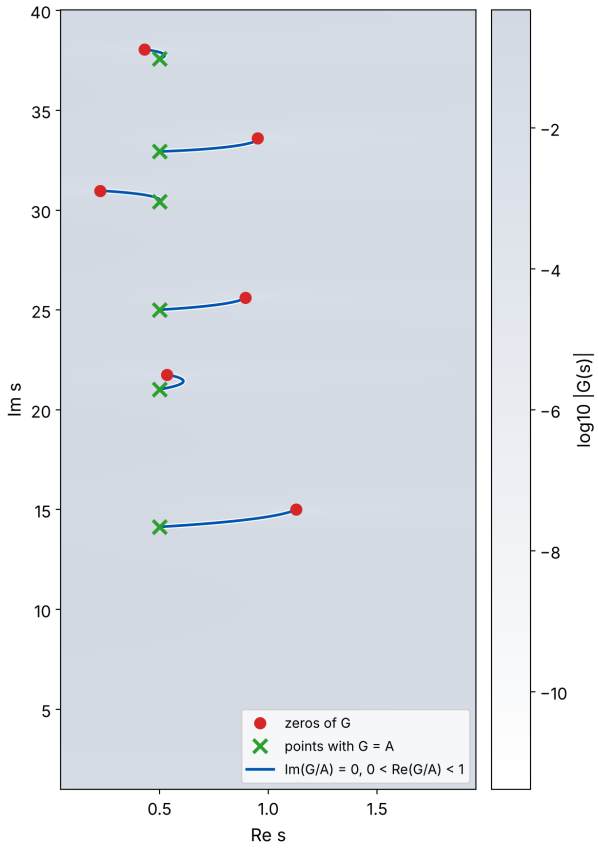


FIGURE 6. Computed zeros of G (dots), points with $G = A$ (crosses), and arcs of $\operatorname{Im}(G/A) = 0$ with $0 < \operatorname{Re}(G/A) < 1$. Each arc joins a zero of G to a point with $G = A$; this pairing is an observation on the computed data.

7. Conclusions

The main conclusion is the bound (4.4). For a two-level probe coupled dispersively to a thermal mode, the geometric mean of the Ramsey contrast after the first return, taken with the weight u^{-2}, satisfies $p_0 < \mathcal{V}_1 < \frac{1}{2}e^{C_g}\,p_0$ at every temperature, where p_0 is the probability that the mode is empty and $C_g = 0.743794\ldots$ is the log-sine constant (3.14). The two constants 1 and $\frac{1}{2}e^{C_g} = 1.05195\ldots$ cannot be improved. The log-sine constant is thus the sharp constant of a measurable quantity: no thermal state gives a ratio $\mathcal{V}_1/p_0$ above $\frac{1}{2}e^{C_g}$, and every smaller constant is exceeded at a sufficiently high temperature.

Five further conclusions follow from the same analysis. The bound holds for every power weight u^{-s-1} with $s > 0$, with the sharp constant $e^{-sG(s)}$, so that the function G on the positive real axis is the logarithm of the sharp constant divided by $-s$ (Theorem 4.1). The occupation is bracketed by the measured geometric contrast alone, $1/\mathcal{V}_1 - 1 < \bar{n} < e^{C_g}/(2\mathcal{V}_1) - 1$, with no fitted quantity (Corollary 4.2). At high temperature $\mathcal{V}_1 = \frac{1}{2}e^{C_g}p_0\,q^{\pi^2/12-3/4}e^{R}$ with $0 < R \le \frac{\tau^2}{\pi^2}(\frac{5}{2} + \frac{1}{2}\log\frac{1}{\tau})$ (Theorem 4.3). The integral of the logarithm of the contrast over all delays with the weight u^{-2} is exactly $-\pi^2\bar{n}$, and the first return separates it into a part within $C_g - \log 2$ of $\log p_0$ and a part that carries the linear term (Proposition 5.1). The product of the contrasts at the harmonic delays N/k decays as $\mathcal{V}_1^N$, with the explicit rate (5.4) (Proposition 5.2).

At complex exponents G is the hot limit of the transform of $\log(V/p_0)$ after the first return, and its zeros are the weights on which that excess has no component (Proposition 6.1). Together with the transform A of the delays before the first return it satisfies $G - A = \frac{\pi}{s}\cot(\frac{\pi s}{2})\zeta$, so that in the strip $0 < \operatorname{Re} s < 2$ the two transforms coincide at a non-real exponent exactly at the zeros of the Riemann zeta function (Proposition 6.2).

These statements concern the ideal model (2.1). They give exact reference values against which a measured sequence of Ramsey contrasts of a thermal mode can be compared.

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